

EVERY EMPTY $3 \times n$ COL BOARD IS A SECOND-PLAYER WIN

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ABSTRACT. Col is the partisan placement game in which two players alternately colour vertices of a graph, a player may never colour a vertex adjacent to one of their own colour, and the player unable to move loses. We prove that for every $n \geq 1$ the empty $3 \times n$ grid has game value 0, so the second player wins. Even widths follow from a half-turn mirror strategy. Odd widths use a one-sided comparison principle that bounds a Col position above by a disjoint sum of small “gadget” positions whenever the cut edges carry no White legality on both sides. Widths $4k + 1$ are covered by a periodic tiling. Widths $4k + 3$ follow by strong induction: three base widths, and a six-case analysis of Blue’s first move in which exactly one case recurses, through a dead separator column, to a strictly narrower board of the same family. The finite inputs are eleven gadget inequalities and three base widths, each certified by a replayable response graph. We give the full written argument, a seam table and a neighbourhood table that make the width-independence checkable by hand, and a description of the search-free verifier. The whole theorem, including every finite certificate, is formalised in Lean 4 without external libraries; the formal proof uses a streamlined window form of the same induction that needs no dyadic numbers.

1. INTRODUCTION

Col was introduced in Conway’s *On Numbers and Games* [2], where it is attributed to Colin Vout, and it is a standard example in combinatorial game theory [1, 4, 3]. Two players, Blue (Left) and White (Right), alternately place a stone of their own colour on an empty vertex of a graph; a player may not place a stone orthogonally adjacent to a stone of the same colour. Under normal play the player who cannot move loses. Figure 1 shows the rule and a position.

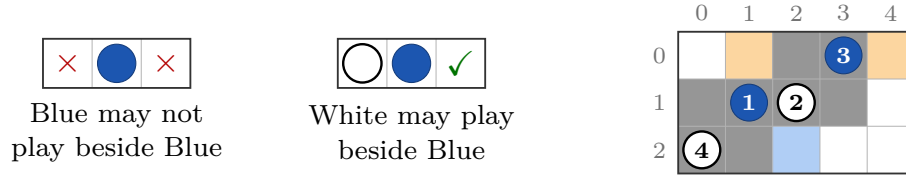


FIGURE 1. Left: the only restriction in Col. Right: a position on the 3×5 board after four moves (numbered). Each empty cell is shaded by who may still play there; see the legend in fig. 3.

Empty rectangular boards are natural test cases. Rectangles with an even side admit a fixed-point-free half-turn and are second-player wins by mirroring. For odd-by-odd rectangles no such symmetry exists, since the centre cell is fixed (fig. 8), and the problem becomes delicate.

Theorem 1.1. *For every integer $n \geq 1$, the empty $3 \times n$ Col board H_n satisfies $H_n = 0$. Equivalently, the second player wins, whoever moves first.*

The proof is computer-assisted in a precise and limited sense. A finite list of local game inequalities (table 1) and three base widths are certified by explicit response graphs that a short

program replays without search. Everything unbounded is proved in the text: the comparison principle, the mirror strategy, colour symmetry, the width-independence of the tilings, and the induction. The finite sweep over widths up to 203 that the verifier also runs is a regression test and plays no logical role. Finally, the entire argument is machine-checked in the Lean 4 proof assistant (section 10), with the finite certificates re-checked inside Lean.

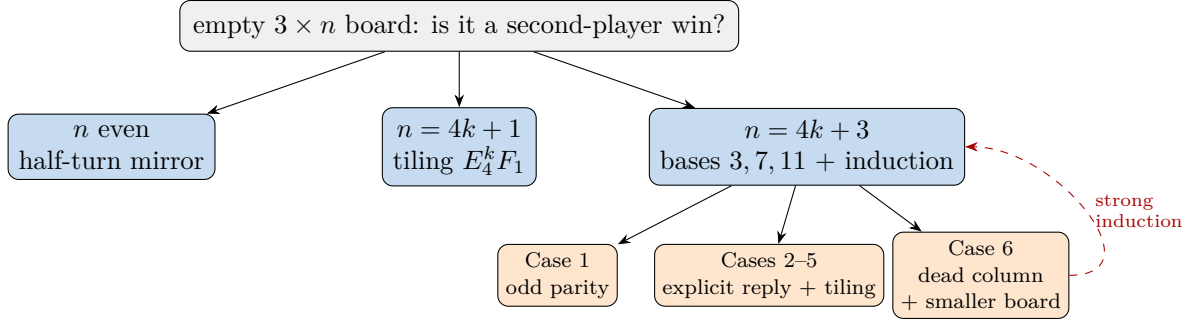


FIGURE 2. Structure of the proof of theorem 1.1.

Organisation. Section 2 fixes conventions. Section 3 proves the comparison principle. Section 4 treats even widths, section 5 the gadgets and their certificates, section 6 widths $4k + 1$, and section 7 widths $4k + 3$. Section 8 proves that the finitely many local checks cover every width. Section 9 describes the certificate verifier, section 10 the Lean formalisation, and section 11 discusses open problems.

2. PRELIMINARIES

2.1. Legality sets. We index cells of the $3 \times n$ board by (r, c) with row $r \in \{0, 1, 2\}$ (top to bottom) and column $c \in \{0, \dots, n - 1\}$. For a cell v let $N[v]$ be v together with its orthogonal neighbours. Since the rule depends on a cell's colour only through legality, a Col position on a graph is determined, for game purposes, by the pair (A, B) of sets of cells legal for Blue and for White. Moves act as follows:

Blue at $v \in A$: $(A, B) \mapsto (A \setminus N[v], B \setminus \{v\})$, White at $w \in B$: $(A, B) \mapsto (A \setminus \{w\}, B \setminus N[w])$.

Any pair (A, B) of vertex sets defines a game by these rules, and we call such a pair a *position* even if no sequence of stones produces it. Gadgets are positions of this kind. We draw positions as *permission diagrams* (fig. 3): each cell is legal for both players, Blue only, White only, or neither. These are permissions, not stones.

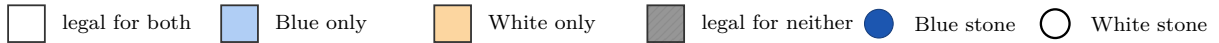


FIGURE 3. Legend for all board figures. Blue is the Left (positive) player.

2.2. Game conventions. We use the standard theory of short partisan games [2, 4]. Blue is Left and White is Right. For games G, H : $G + H$ is the disjoint sum, $-G$ swaps the roles, and $G \leq H$ means that Right, moving second, wins $H - G$. In particular $G \leq 0$ means Blue loses moving first; $G < 0$ means White wins whoever moves first; $G = 0$ means the second player wins. We write $G \triangleleft 0$ for “not $0 \leq G$ ”, that is, White moving first wins. Recall that $G \leq 0$ iff every Left option G^L satisfies $G^L \triangleleft 0$, and $G^L \triangleleft 0$ holds if G^L has a Right option $G^{LR} \leq 0$ or if $G^L < 0$. Dyadic rationals are games in the usual way, and $1 = \{0 \mid \}$, $\frac{1}{4} = \{0 \mid \frac{1}{2}\}$.

A Col position on a disjoint union of graphs is the disjoint sum of its restrictions. Swapping every Blue permission with every White permission sends the position (A, B) to (B, A) , which is its negative. The empty board has $A = B$ and so equals its own negative:

Lemma 2.1 (colour symmetry). $H_n = -H_n$. Hence $H_n \leq 0$ implies $H_n = 0$.

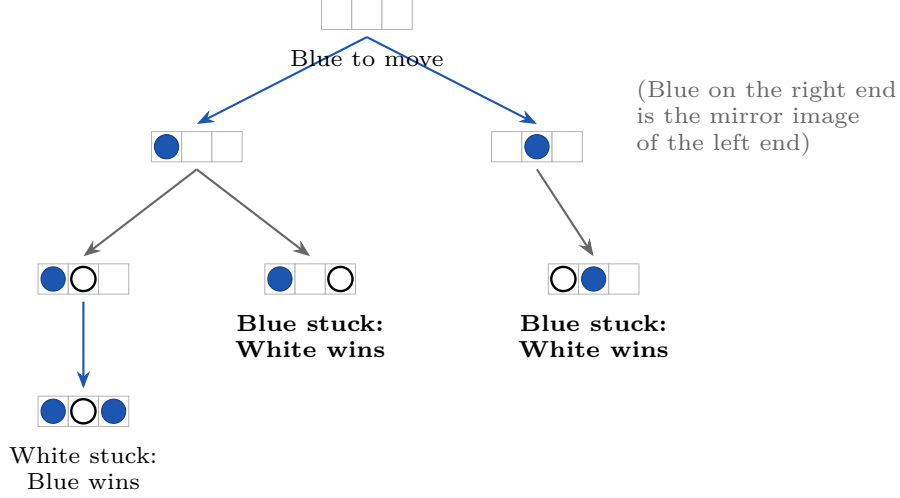


FIGURE 4. The complete Blue-first game tree of the 1×3 strip, up to mirror symmetry. Every Blue opening has a White reply after which Blue is stuck. The strip is also a second-player win when White moves first, so it has value 0; this is the case $n = 1$ of theorem 1.1.

3. THE COMPARISON PRINCIPLE

The central tool compares a position with a sum of smaller positions obtained by cutting the board, provided the cut cannot help White.

Definition 3.1. Let Γ be a graph and $P = (A, B)$ a position on Γ . A *virtual decomposition* of P consists of a partition of the vertices into blocks, with the virtual graph $\Gamma^* \subseteq \Gamma$ obtained by deleting every edge between different blocks, and a position $T = (A^*, B^*)$ on Γ^* , such that

- (C1) $A \subseteq A^*$ (Blue keeps every move),
- (C2) $B^* \subseteq B$ (White gains no move), and
- (C3) no deleted edge has both endpoints in B^* .

We write $T = \sum_i T_i$ for the resulting disjoint sum of block positions.

Theorem 3.2 (comparison principle). *If T is a virtual decomposition of P , then $P \leq T$.*

Proof. We show that Blue, moving first, loses $P + (-T)$. In $-T$ the players' roles are exchanged: a Blue move of $-T$ is a White placement in T , and a White move of $-T$ is a Blue placement in T . White keeps the invariant that the current positions $P = (A, B)$ and $T = (A^*, B^*)$ satisfy (C1)–(C3).

If Blue plays $v \in A$ in P , then $v \in A^*$ by (C1), and White answers in $-T$ with the Blue placement at v in T . Afterwards $A \setminus N_\Gamma[v] \subseteq A^* \setminus N_{\Gamma^*}[v]$ because $N_{\Gamma^*}[v] \subseteq N_\Gamma[v]$, and $B^* \setminus \{v\} \subseteq B \setminus \{v\}$. Condition (C3) persists because B^* only shrank.

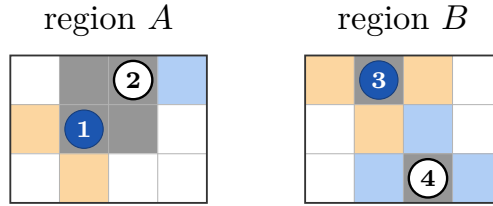
If Blue plays in $-T$, this is a White placement at some $w \in B^*$, and White answers by playing w in P , which is legal by (C2). Then $A \setminus \{w\} \subseteq A^* \setminus \{w\}$. For (C2), let $u \in B^* \setminus N_{\Gamma^*}[w]$. Then

$u \in B$, and $u \notin N_\Gamma[w]$: otherwise uw is a deleted edge with both endpoints in B^* , contradicting (C3). Hence $u \in B \setminus N_\Gamma[w]$. Condition (C3) persists.

White therefore always has a reply. The game is finite, so Blue eventually cannot move and loses. Hence $P - T \leq 0$. $\square$

Corollary 3.3. *Let $T = \sum_i T_i$ be a virtual decomposition of P and suppose $T_i \leq b_i$ for dyadic rationals b_i . If $\sum b_i \leq 0$ then Blue, moving first, loses P . If $\sum b_i < 0$ then $P < 0$, so White also wins moving first.*

Proof. By theorem 3.2 and monotonicity of addition, $P \leq T \leq \sum b_i$. $\square$



White always answers in the same region as Blue.
If White wins each region as second player, she wins the sum.

FIGURE 5. Why sums of nonpositive games are nonpositive: White replies in the component Blue just played in.

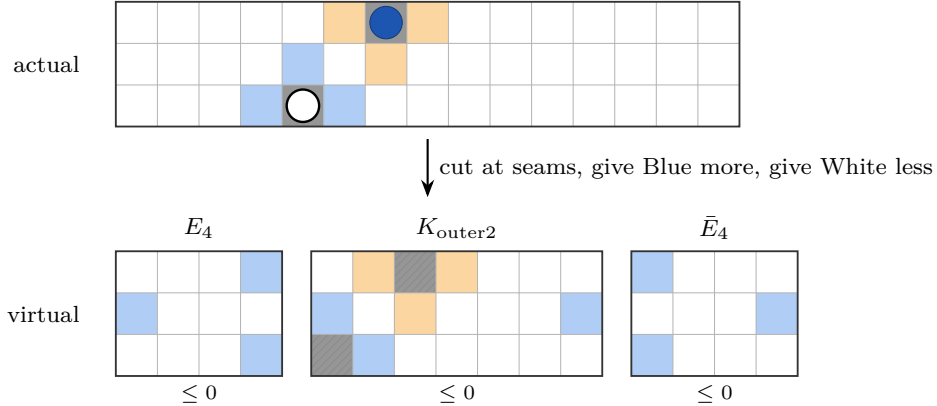


FIGURE 6. A virtual decomposition, from Case 3 of section 7 on 3×15 . Top: the actual position after Blue's opening and White's reply. Bottom: the three virtual blocks. Each block may give Blue extra permissions and White fewer, and at each red seam at most one side is White-legal in every row.

Remark 3.4. Theorem 3.2 is one-sided. Cutting edges does not in general preserve value, and (C3) is essential: an edge with White-legal endpoints on both sides is a restriction on White that the virtual game would forget.

3.1. Seams and ports. In all our decompositions the blocks are intervals of consecutive columns, so only horizontal edges between the last column of one block and the first column of the next are deleted. For a block position, its *left and right White ports* are the White-legality patterns of its first and last columns, written top to bottom as binary words. Condition (C3) at a seam says exactly that the right port of the left block and the left port of the right block have disjoint supports (fig. 7).

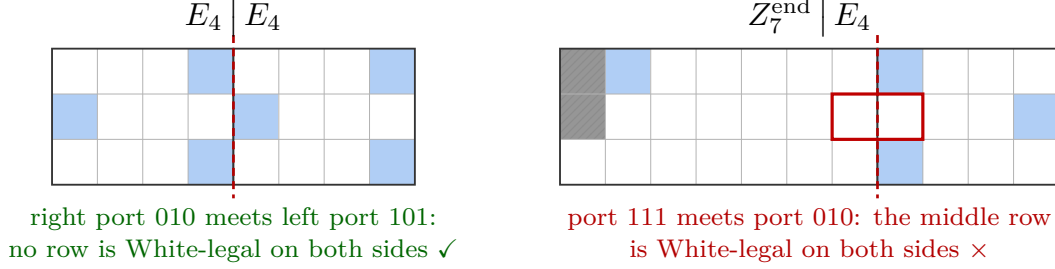


FIGURE 7. A compatible seam (left) and an incompatible one (right).

4. EVEN WIDTHS

Proposition 4.1. *If n is even then $H_n = 0$.*

Proof. The half-turn $\sigma(r, c) = (2 - r, n - 1 - c)$ is a graph automorphism of the $3 \times n$ grid with no fixed cell, since a fixed cell would need $c = (n - 1)/2$. Let the second player answer every move at v by the move at $\sigma(v)$. We claim that after each answer the position satisfies $B = \sigma(A)$ (for A, B the Blue and White legal sets, when Blue moved first; symmetrically otherwise). This holds initially. If Blue plays $v \in A$ then $\sigma(v) \in B$, and $\sigma(v) \neq v$, so $\sigma(v)$ is still White-legal after Blue's move. After both moves the new sets are $A' = A \setminus (N[v] \cup \{\sigma v\})$ and $B' = B \setminus (\{v\} \cup N[\sigma v])$, and $\sigma(A') = B'$ because σ maps $N[v]$ to $N[\sigma v]$ and is an involution. The second player always has a reply and so wins. Note that v and $\sigma(v)$ may be adjacent; this is harmless because they receive different colours. $\square$

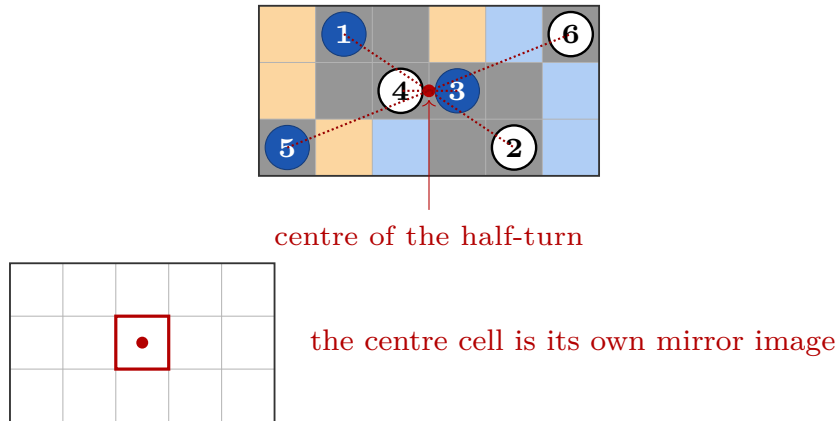


FIGURE 8. Left: the half-turn strategy on 3×6 ; stones 2, 4, 6 are White's rotated answers to 1, 3, 5. Right: on odd widths the centre cell is fixed, and the strategy breaks down.

5. GADGETS AND THEIR CERTIFICATES

5.1. The gadgets. A *gadget* is a position on a $3 \times w$ board, given by Blue and White legality masks. We encode a mask by the integer $\sum 2^{r+w+c}$ over its legal cells. Table 1 lists the gadgets and bounds used; figs. 9 and 10 draw them. Gadgets with a local move are the local positions after Blue’s opening and White’s reply, with White possibly forgoing further permissions.

Gadget	w	Blue mask	White mask	ports L/R	bound	used in
E_4	4	4095	2023	101/010	≤ 0	$4k + 1$; Cases 1, 3, 4, 5
$\bar{E}_4$	4	4095	3710	010/101	≤ 0	Cases 2, 3, 4, 6
F_1	1	7	5	101/101	≤ 0	$4k + 1$
K_{corner}	3	244	94	011/100	≤ 0	Case 2
K_{outer2}	7	2080241	2039675	100/101	≤ 0	Case 3
K_{outer4}	7	1046471	515951	101/100	≤ 0	Case 4
T_{middle1}	6	253500	130844	001/010	≤ 0	Case 5
J_5	5	32767	21845	101/101	≤ 0	Case 5
Z_7	7	2097022	2088828	001/101	≤ 0	Case 6
C_3	3	511	503	101/111	$\leq \frac{1}{4}$	Case 1
$E_4^{(r,c)}$, $r + c$ odd	4	$4095 \setminus N[v]$, $2023 \setminus \{v\}$		$\subseteq 101/010$	≤ -1	Case 1

TABLE 1. The gadget inequalities. K_{outer4} is the left–right reflection of K_{outer2} . The row-2 opened E_4 bounds are the top–bottom reflections of the row-0 ones. A second version of Z_7 (White mask 2097020, right port 111) appears only in the stored base assemblies, at a physical board edge.

5.2. Certificates. A claim $G + q \leq 0$, with G a gadget and q a dyadic rational, is certified by a *response graph*: a finite set $\mathcal{S}$ of positions of the form $X + q'$, containing $G + q$, together with a map that assigns to every Left option of every member of $\mathcal{S}$ a Right option of that option lying in $\mathcal{S}$ (or a terminal position in which Left has no move). By induction on the finite game, every member of $\mathcal{S}$ is ≤ 0 . Checking a response graph requires only generating Left options and testing membership; no search is performed. The bound $C_3 \leq \frac{1}{4}$ is certified as $C_3 - \frac{1}{4} \leq 0$ and $E_4^{(r,c)} \leq -1$ as $E_4^{(r,c)} + 1 \leq 0$.

6. WIDTHS CONGRUENT TO 1 MODULO 4

Proposition 6.1. $H_{4k+1} = 0$ for every $k \geq 0$.

Proof. Decompose the empty board into k copies of E_4 followed by F_1 (fig. 11). (C1) holds because every block is Blue-legal everywhere, and (C2) trivially. The seams are $E_4|E_4$ (010 against 101) and $E_4|F_1$ (010 against 101), so (C3) holds. By corollary 3.3, $H_{4k+1} \leq 0$, and lemma 2.1 gives equality. $\square$

7. WIDTHS CONGRUENT TO 3 MODULO 4

7.1. Base cases. The boards H_3 , H_7 , H_{11} are certified directly. For each of their 4, 8 and 12 openings up to symmetry, the data give a White reply and a virtual decomposition by certified tiles (the gadgets above and nine further ordinary tiles), checked by the same comparison conditions. Figure 12 shows the replies on 3×3 . The six-case step below does not apply to these widths: at $n = 7$ it would fail at $(1, 1)$ and $(1, 3)$, and at $n = 11$ at $(1, 5)$.

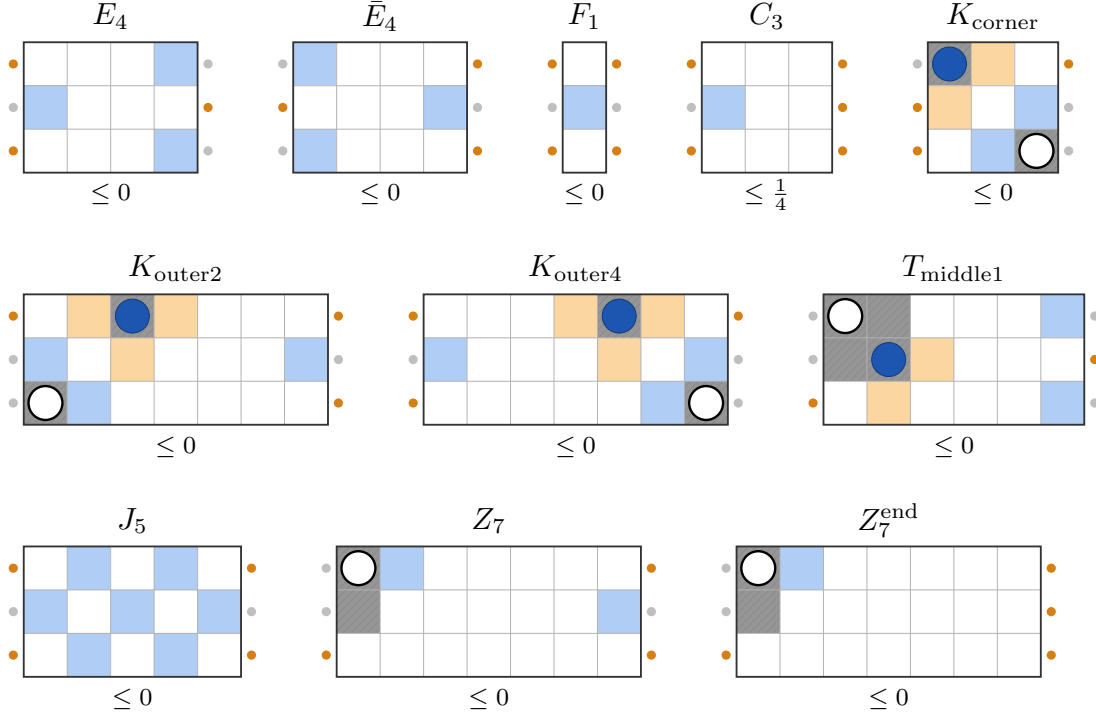


FIGURE 9. The gadgets, with their certified bounds. Dots beside each gadget show its White ports (orange: White-legal). Stones mark the local moves already played. For Z_7 Blue's stone lies in the column just to its left.

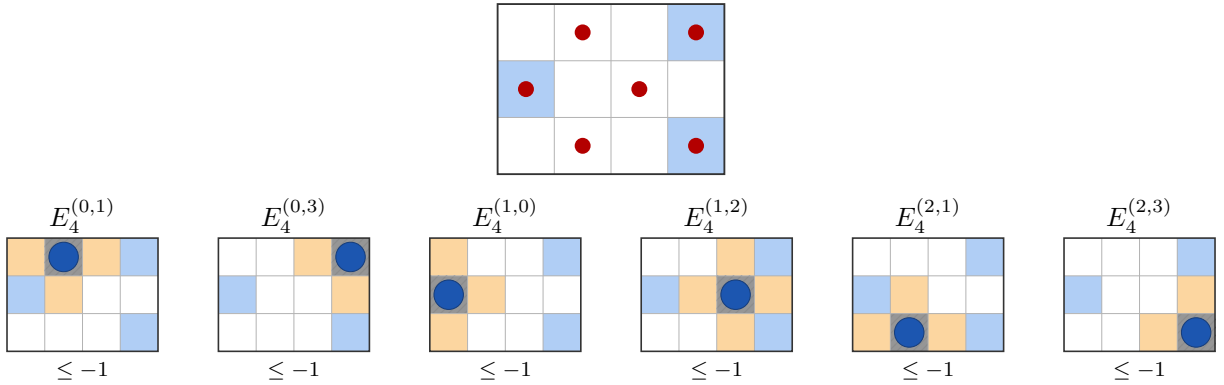
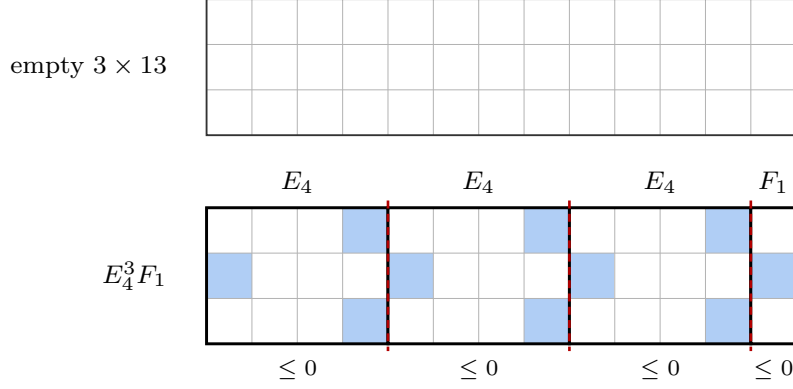
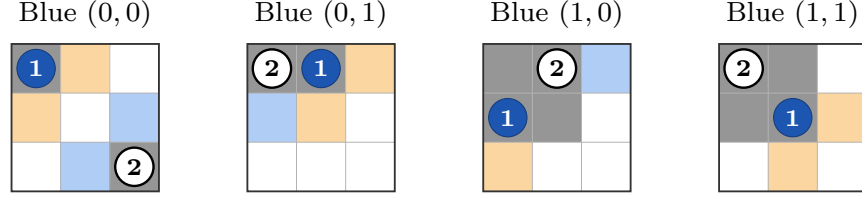


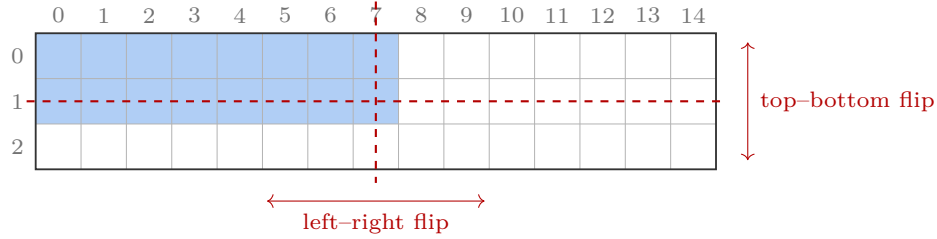
FIGURE 10. Top: the six odd-parity cells of E_4 (dots), those with $r + c$ odd. Bottom: after Blue opens any of them, the remaining position is at most -1 .

7.2. The inductive step. Let $n = 4k + 3$ with $k \geq 3$, and assume $H_m = 0$ for every $m < n$ with $m \equiv 3 \pmod{4}$. By lemma 2.1 it suffices to show $H_n \leq 0$, that is, $H_n^L \triangleleft 0$ for every Blue opening H_n^L .

Normalisation. The top-bottom and left-right reflections are automorphisms of the board that preserve checkerboard parity $r + c \pmod{2}$ (the latter because $n - 1 = 4k + 2$ is even). We may therefore assume that Blue opens at (r, c) with $r \in \{0, 1\}$ and $0 \leq c \leq 2k + 1$ (fig. 13). Every such opening falls in exactly one of six cases (fig. 14): odd $r + c$ (Case 1); $r = 0$ and $c = 0$ (Case

FIGURE 11. The tiling $E_4^k F_1$, here for $n = 13$.FIGURE 12. The four openings of 3×3 up to symmetry and White's certified replies.

2), $c \equiv 2$ (Case 3), or $c \equiv 0, c \geq 4$ (Case 4) modulo 4; $r = 1$ and $c \equiv 1$ (Case 5) or $c \equiv 3$ (Case 6) modulo 4.

FIGURE 13. Normalisation on 3×15 : by the two reflections every opening is equivalent to one in the shaded region.FIGURE 14. The case used for each normalised opening on 3×23 .

In each case we give a decomposition of the position and apply corollary 3.3; section 8 verifies conditions (C1)–(C3) for all k . Figures 15–20 show each case on 3×19 ($k = 4$). In every figure the top board is the actual position and the bottom board the virtual decomposition, with seams dashed.

Case 1: $r + c$ odd. Use $E_4^k C_3$ with no White reply. Since $c \leq 2k + 1 < 4k$, Blue opened block $j = \lfloor c/4 \rfloor$ at the local cell $(r, c - 4j)$, which has odd parity because $4j$ is even. With $r \in \{0, 1\}$ this local cell is one of $(0, 1), (0, 3), (1, 0), (1, 2)$, and that block becomes $E_4^{(r, c-4j)} \leq -1$. The other E_4 blocks are ≤ 0 and $C_3 \leq \frac{1}{4}$, so the position is at most $-1 + \frac{1}{4} = -\frac{3}{4} < 0$ (fig. 16), and White, to move, wins.

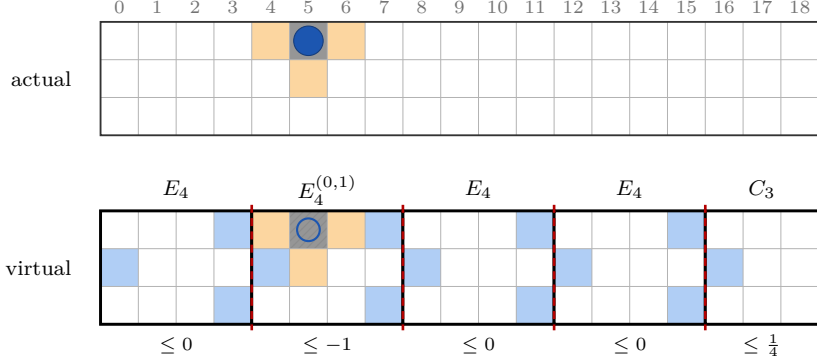

 FIGURE 15. Case 1, Blue at $(0, 5)$ on 3×19 .


FIGURE 16. The bookkeeping in Case 1.

Case 2: corner $(0, 0)$. White replies at $(2, 2)$. Decompose as $K_{\text{corner}} \bar{E}_4^k$; the widths sum to $3 + 4k = n$.

Case 3: $r = 0, c = 4a + 2$. White replies at $(2, c - 2)$. Decompose as $E_4^a K_{\text{outer2}} \bar{E}_4^b$ with $b = k - a - 1$; the widths sum to $4a + 7 + 4b = n$. From $4a + 2 \leq 2k + 1$ we get $b \geq (2k - 3)/4 > 0$, so $b \geq 1$.

Case 4: $r = 0, c = 4a + 4$. White replies at $(2, c + 2)$. Decompose as $E_4^a K_{\text{outer4}} \bar{E}_4^b$ with $b = k - a - 1$. Here $a \geq 0$, and $4a + 4 \leq 2k + 1$ gives $b \geq (2k - 1)/4 > 0$, so $b \geq 1$.

Case 5: $r = 1, c = 4a + 1$. White replies at $(0, c - 1)$. Decompose as $E_4^a T_{\text{middle1}} E_4^b J_5$ with $b = k - a - 2$; the widths sum to $4a + 6 + 4b + 5 = n$. From $c \leq 2k + 1, a \leq \lfloor k/2 \rfloor \leq k - 2$ for $k \geq 3$, so $b \geq 0$.

In Cases 2–5 every block is ≤ 0 , so after White's reply Blue, to move, loses by corollary 3.3. Hence H_n^L has a Right option ≤ 0 .

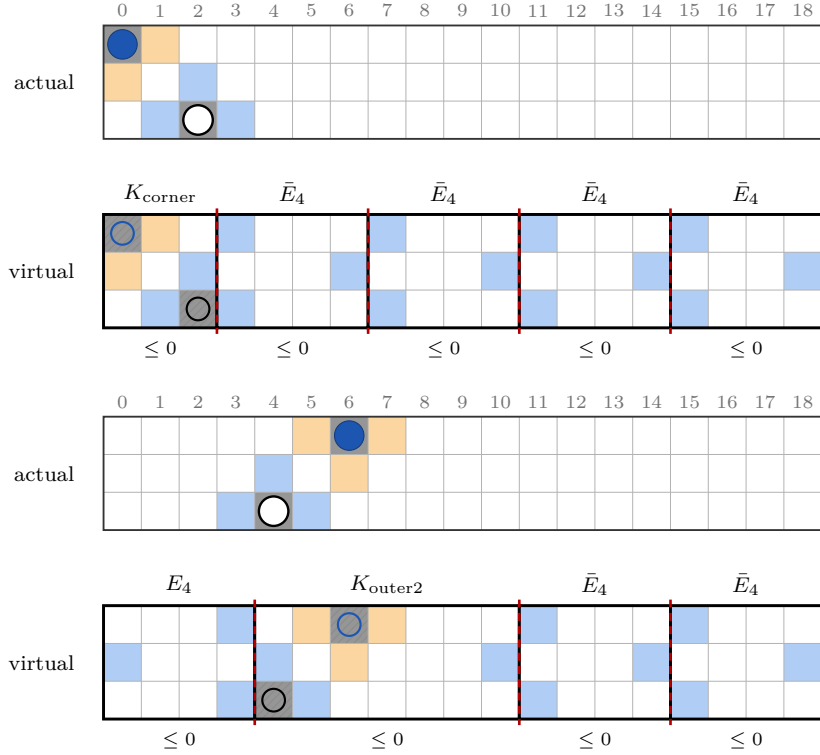


FIGURE 17. Case 2, Blue at $(0,0)$; Case 3, Blue at $(0,6)$ ($a = 1, b = 2$).

Case 6: $r = 1, c \equiv 3 \pmod{4}$ (the recursive case). White replies at $(0, c+1)$. Blue's stone at $(1, c)$ makes all three cells of column c Blue-illegal. White's reply makes $(0, c)$ White-illegal, and White voluntarily gives up $(2, c)$, which is allowed by (C2). Column c is then a *dead separator* (fig. 19): it carries no permissions in the virtual position, so it contributes the game 0, and no deleted edge at it has a White-legal endpoint in that column.

On the left, restore every Blue permission lost in columns $0, \dots, c-1$, which is allowed by (C1). The left block is then exactly the empty board H_c . On the right the width is $s = n - c - 1 \equiv 3 \pmod{4}$, with $s \geq 4k + 2 - (2k + 1) \geq 7$, and we decompose it as $Z_7 \bar{E}_4^{(s-7)/4}$. Since $3 \leq c < n$ and $c \equiv 3 \pmod{4}$, either $c \in \{3, 7, 11\}$ (base cases) or $c \geq 15$ and the induction hypothesis applies. Either way $H_c = 0$, and

$$\text{position after White's reply} \leq H_c + 0 + Z_7 + \sum \bar{E}_4 \leq 0.$$

Conclusion of the step. Every normalised opening is covered. In Cases 2–6 White's stated reply leads to a position ≤ 0 . In Case 1 the position after Blue's move is already negative. Thus $H_n \leq 0$, and $H_n = 0$ by lemma 2.1. Together with propositions 4.1 and 6.1, this proves theorem 1.1.

8. WIDTH INDEPENDENCE

The decompositions above depend on k, a and b . We show that conditions (C1)–(C3) reduce to finitely many checks independent of these parameters.

Lemma 8.1 (locality). *In each case, (C1)–(C3) hold for all admissible parameters provided that:*

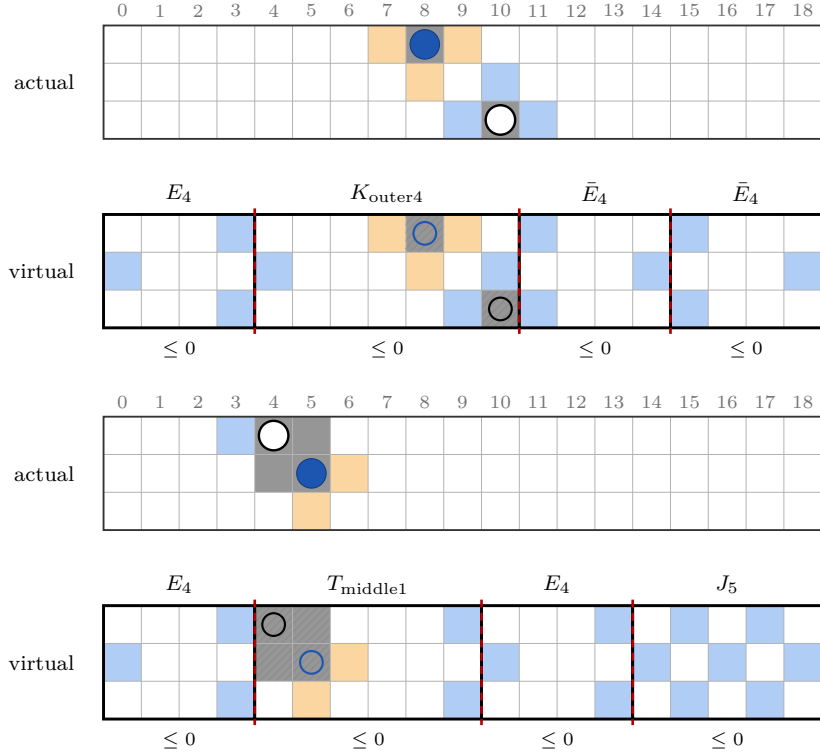
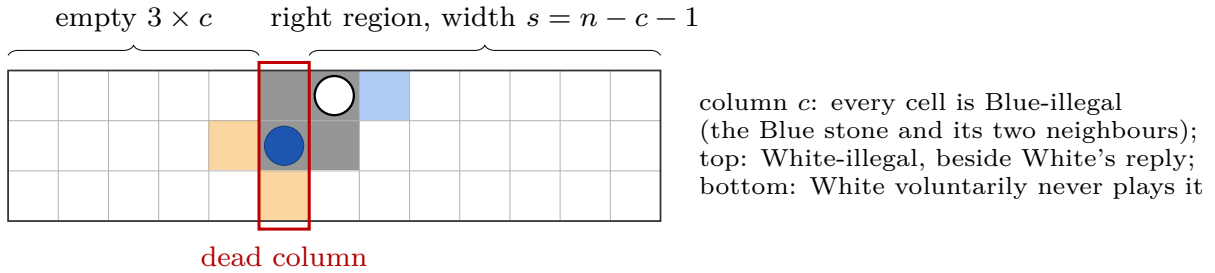

 FIGURE 18. Case 4, Blue at $(0, 8)$ ($a = 1, b = 2$); Case 5, Blue at $(1, 5)$ ($a = 1, b = 1$).


FIGURE 19. The dead separator column of Case 6.

- (i) every block not containing the opening or the reply has full Blue mask;
- (ii) the exceptional block's masks satisfy (C1) and (C2) against the actual local position; and
- (iii) every seam that occurs is compatible, and every cell of $N[w]$ for the White reply w that lies outside the exceptional block is White-illegal in the virtual position.

Proof. The actual position differs from the empty board only in the cells $N[v] \cup \{w\}$, removed from A , and $\{v\} \cup N[w]$, removed from B (in Case 1 there is no w). For a block other than the exceptional one, (C1) holds by (i). For (C2), a virtual White cell of such a block is White-legal in the actual position unless it equals v or lies in $N[w]$. The cell v lies in the exceptional block, and the cells of $N[w]$ outside it are White-illegal virtually by (iii). The exceptional block is (ii), and (C3) is the seam part of (iii). $\square$

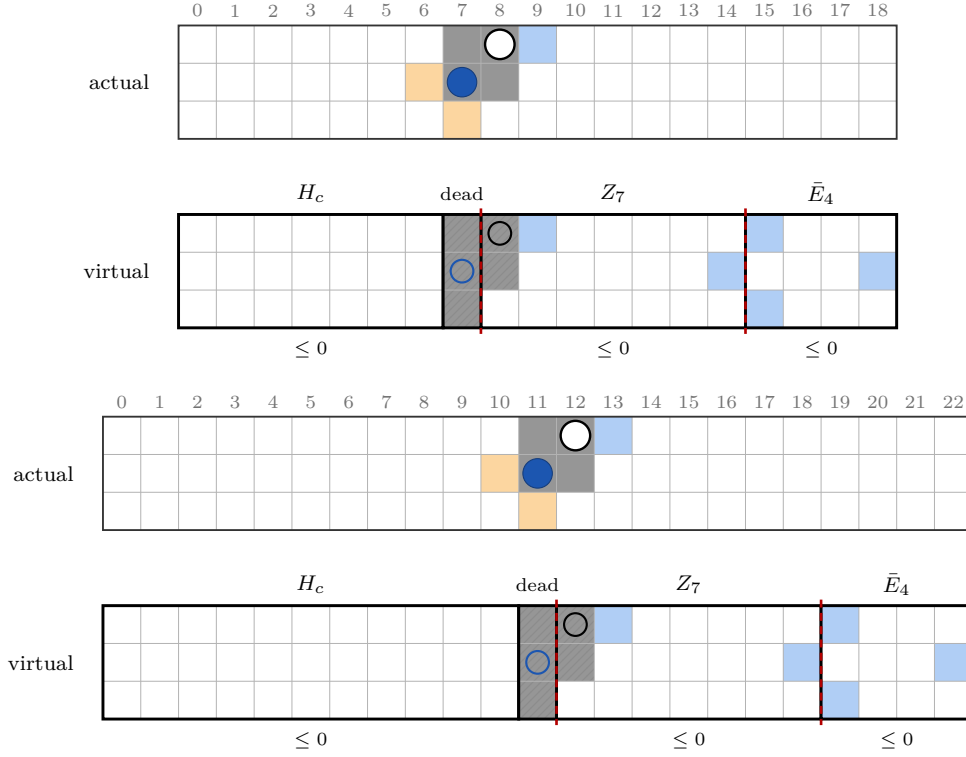


FIGURE 20. Case 6. Top: Blue at $(1, 7)$ on 3×19 recurses to H_7 . Bottom: Blue at $(1, 11)$ on 3×23 recurses to H_{11} .

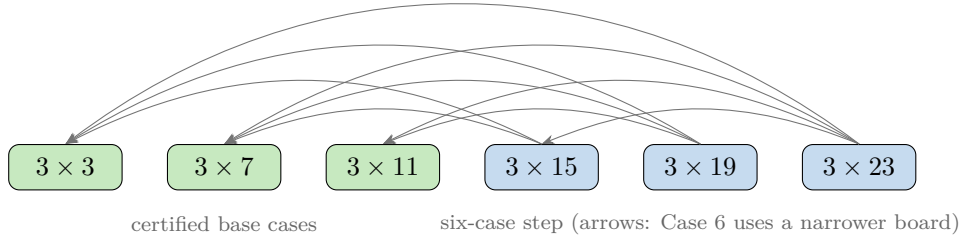


FIGURE 21. Dependencies among the widths $n \equiv 3 \pmod{4}$. Each arrow is a Case 6 call to a narrower board; only the widths 3, 7, 11 are certified directly.

Condition (i) holds because E_4 , $\bar{E}_4$, F_1 , J_5 and C_3 have full Blue masks, and the left block of Case 6 is the empty H_c . Condition (ii) is a finite check for each of the gadgets with local moves. It is satisfied by construction, since each gadget's root is the local position after the moves with some White permissions removed. Tables 2 and 3 list every seam and every cross-block neighbour of a White reply.

9. VERIFICATION

The package in `proofs/construction/three_row` of the accompanying repository contains the certificates and a single standard-library Python verifier, `verify.py`. It performs no game search and runs in a few seconds. It checks:

Seam (left right)	ports	where
$E_4 E_4, E_4 F_1, E_4 J_5, E_4 C_3$	010 101	$4k + 1$; Cases 1, 5
$E_4^{(r,c)} E_4$ or C_3	$\subseteq$ 010 101	Case 1
$E_4 E_4^{(r,c)}$	010 $\subseteq$ 101	Case 1
$\bar{E}_4 \bar{E}_4$	101 010	Cases 2, 3, 4, 6
$K_{\text{corner}} \bar{E}_4$	100 010	Case 2
$E_4 K_{\text{outer2}}, K_{\text{outer2}} \bar{E}_4$	010 100, 101 010	Case 3
$E_4 K_{\text{outer4}}, K_{\text{outer4}} \bar{E}_4$	010 101, 100 010	Case 4
$E_4 T_{\text{middle1}}, T_{\text{middle1}} E_4$ or J_5	010 001, 010 101	Case 5
$H_c \text{dead} Z_7, Z_7 \bar{E}_4$	$\cdot$ 000 $\cdot$, 101 010	Case 6

 TABLE 2. Every seam in the six cases and the $4k + 1$ tiling. In each row the two ports have disjoint supports.

Case	reply w (local)	neighbour outside block	White-illegal because
2	(2, 2) in K_{corner}	(2, 3), first column of $\bar{E}_4$	left port 010
3	(2, 0) in K_{outer2}	(2, $4a - 1$), last column of E_4 ($a \geq 1$)	right port 010
4	(2, 6) in K_{outer4}	(2, $4a + 7$), first column of $\bar{E}_4$	left port 010
5	(0, 0) in T_{middle1}	(0, $4a - 1$), last column of E_4 ($a \geq 1$)	right port 010
6	(0, 0) in Z_7	(0, c), dead column	dead

TABLE 3. Neighbours of White's reply lying outside the exceptional block.

- (1) SHA-256 hashes of the 34 certificate files;
- (2) a search-free replay of 21 ordinary response graphs (20,575 checkpoints, 82,899 edges) and 12 dyadic ones (1,064 checkpoints, 3,130 edges), including reflections rebuilt from their sources;
- (3) the binding of each of the 17 gadget roles to its exact root, ports and bound;
- (4) the stored base assemblies for widths 3, 7, 11;
- (5) as a regression test, all 6,651 assemblies for widths up to 203, including 5,280 normalised openings of the six-case step;
- (6) the local windows (13 plus 2 supplementary row-2 windows) and the 25 seam types, all of which already occur by $n = 19$.

A separate script applies 15 targeted corruptions, such as a wrong reply, a flipped port bit, or a weakened C_3 bound. The verifier rejects every one for the expected reason.

10. FORMAL VERIFICATION IN LEAN

The theorem is formalised in Lean 4, using only Lean's core library. The development is in the directory `proofs/lean` of the accompanying repository. The final statement is phrased with the literal rules of Col:

```
theorem three_by_n_stones (n : Nat) :
  StoneLoses (board n) (fun _ => False) (fun _ => False)
```

Here `StoneLoses R M 0` means that the player to move, owning the stones `M`, loses against the owner of the stones `0`, where a placement is legal if the cell of `R` is empty and not orthogonally adjacent to a stone of the mover's colour. Starting from no stones, it states that the first player loses on every empty $3 \times n$ board, whichever colour starts. The development consists of five modules.

Game:

The legality-set model and an inductive predicate **Loses**, whose proofs are finite winning strategies for the second player. It proves the comparison principle (theorem 3.2, for an arbitrary subgraph and cut condition), invariance under relabelling of cells, and the sum rule for non-adjacent regions.

Cert:

An encoding of $3 \times w$ positions as natural numbers, a search-free checker for closed certificate sets, and its soundness theorem: every position in an accepted set is a loss for the player to move.

Blocks:

Seams between blocks, repeated E_4 and $\bar{E}_4$ bricks, and a general *window step* that turns a local domination check around Blue's opening and White's reply into a proof of the whole position. It is the locality lemma (lemma 8.1) in general form.

Main:

The mirror strategy, the reflections used for normalisation, the $E_4^k F_1$ tiling, the six cases, and the strong induction.

Stones:

The equivalence between the stone rules and the legality-set model.

A streamlined induction. The formal proof follows section 7 but handles all six cases in one shape (fig. 22): the position after Blue's opening and White's reply is compared with $E_4^a W \bar{E}_4^b$, and in Case 6 with $H_c W \bar{E}_4^b$. The window W is simply the actual local position after the two moves. White gives up only the seam cells that the neighbouring ports require: the middle cell of each edge column next to an E_4 or $\bar{E}_4$, and the whole first column next to H_c (fig. 23, table 4). Two changes remove all numerical bookkeeping:

- In Case 1, White replies explicitly at local $(1, 4)$ of a seven-column window starting at the opened brick. The bound $-1 + \frac{1}{4} < 0$ is not needed.
- In Case 5, a seven-column window replaces $T_{\text{middle1}} E_4^b J_5$.

With these windows the step already holds for $n = 11$ (Case 6 then recurses to H_3), so the only base cases are the whole boards 3×3 and 3×7 . The even widths and the $4k + 1$ tiling are formalised exactly as in sections 4 and 6.

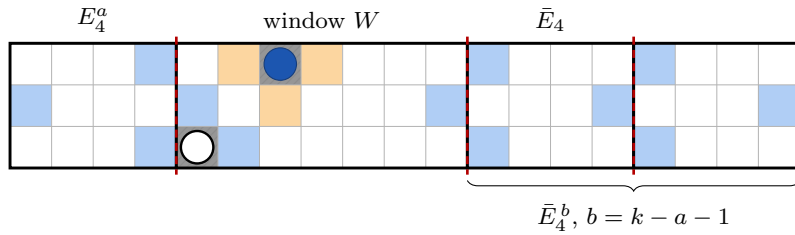


FIGURE 22. The shape of every case in the Lean proof, shown for Case 3 on 3×19 : bricks E_4^a , one window containing both moves, and bricks $\bar{E}_4^b$.

Trusted base. Everything except the 14 certificate checks is verified by Lean's kernel. The certificate checks are evaluated by compiled code through Lean's `native_decide`, so the trusted base consists of the Lean kernel, the Lean compiler, and the axioms `propext`, `Classical.choice` and `Quot.sound`. The certificates were produced by a separate exhaustive search, which is not trusted: soundness depends only on the checker. The build takes about 20 seconds on a laptop.

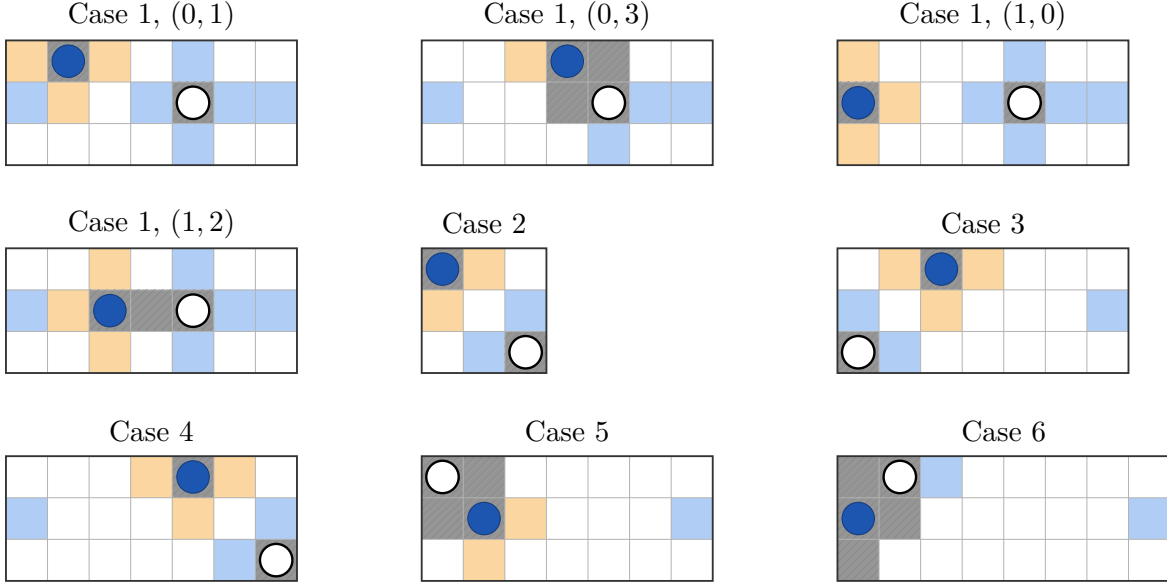


FIGURE 23. The nine window gadgets of the Lean proof. Each is the actual local position after Blue's opening and White's reply, minus the seam cells White gives up.

Window	width	opening (local)	reply (local)	left neighbour	checkpoints
Case 1	7	$(0, 1), (0, 3), (1, 0), (1, 2)$	$(1, 4)$	E_4^a	1456, 1056, 1314, 910
Case 2	3	$(0, 0)$	$(2, 2)$	board edge	7
Case 3	7	$(0, 2)$	$(2, 0)$	E_4^a	1478
Case 4	7	$(0, 4)$	$(2, 6)$	E_4^a	1253
Case 5	7	$(1, 1)$	$(0, 0)$	E_4^a	891
Case 6	8	$(1, 0)$	$(0, 1)$	H_c	4844
$E_4, \bar{E}_4, F_1$	4, 4, 1				152, 155, 3
H_3, H_7	3, 7				40, 9819

TABLE 4. The certificates checked inside Lean. The right neighbour of every window is $\bar{E}_4^b$.

11. DISCUSSION

The recursive Case 6 is the heart of the argument. A single middle-row stone, together with a reply and one voluntary concession by White, isolates an ordinary empty board, which reduces the problem to a narrower width in the same residue class. We do not know an analogue for five rows. There a single stone does not kill a column, and experiments suggest that fixed boundary states are not numerically stable when neutral columns are inserted. The cases $5 \times n$ for odd n , and general odd-by-odd rectangles, remain open.

Verification status. Every part of the proof is machine-checked: the certificates by the Python verifier of section 9, and the complete theorem, including its certificates, by the Lean formalisation of section 10.

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